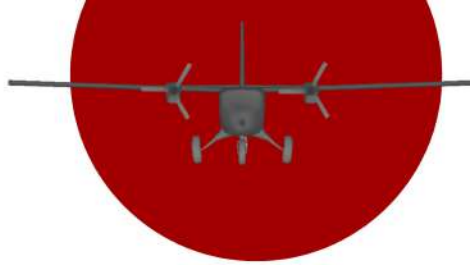
AEROMBULANCE : THE AEROEVACUATION MEDICAL AIRCRAFT

Basant Karki, Abinash Kumar Yadav, Arun Panta, Prabin Baniya, Pranish Biswokarma Sameer Baskota

IOE, Pulchowk Campus



Introduction

Aerombulace is aeromedical evacuation aircraft which is especially designed for transporting fatal injured or critically illed patients. Its primary purpose is to provide rapid medical evacuation and in-flight medical care, particularly in situations where ground ambulances are too slow or unable to reach the location. Air ambulances typically carry medical personnel, life-support equipment, and patient monitoring systems, enabling continued treatment during transit. They are commonly used to access remote or difficult-to-reach areas, and for long-distance transfers where time is critical.

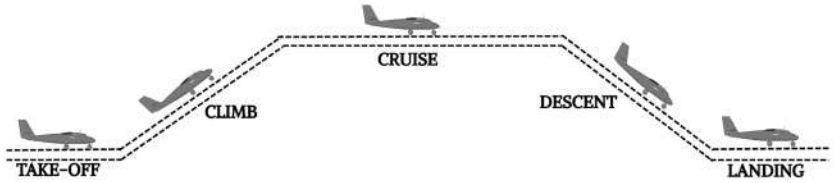
Design Requirements

Aeromedical Evacuation Aircraft
Capable of operating at length runaway
Climb to cruise altitude 160 knots
Two stretcher and one NICU (Neo Natal Intensive Care Unit)
STOL (Short takeoff and Landing AIrcraft) operate at upper hilly regions.

Aircraft Configuration

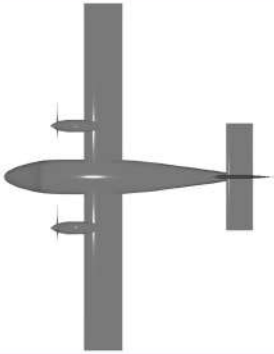
- Twin-engine turboprop (PT6A-20): Reliable and capable of short-field operations for remote runways.
- High wing: Good visibility and clearance for patient/equipment handling.
- STOL performance: Access to short, rough strips, reducing evacuation time.
- Spacious cabin: Configurable for stretchers, crew, and medical systems.
- Multi-role adaptability: Quick change between passenger, cargo, and ambulance missions with medical equipment support.
- Operational versatility: Effective in both IFR/VFR and challenging terrains like mountains and remote areas.

MISSION PROFILE

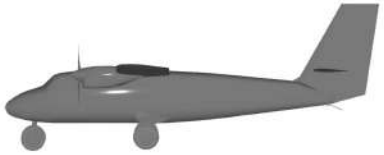




Isometric View



Top View

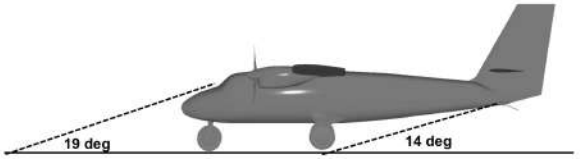


Side View



Front View

PILOT LOS & TIP-BACK ANGLE



INTERNAL LAYOUT



SPECIFICATIONS

PARAMETERS	Values
MTOW	5200 kg
Range	1200 km
Endurance	5.1 hours
Stall Speed	83 knots
Cruise Speed	160 knots
Maximum Speed	170 knots
Absolute Ceiling	7620 m
Payload Weight	1300 kg
Empty Weight	3300 kg
Fuel Capacity	600 kg
Powerplant	PT6A-20
Power	450 HP
Fuselage Length	15.7m

WING & TAIL SIZING

SURFACE	WING	HORIZONTAL TAIL	VERTICALL TAIL
Airfoil	NACA 4416	NACA 0009	NACA 0009
Span (m)	19.26	11.87	16.09
Area (m^2)	41.09	17.8	18.39
AR	9.03	7.01	14.08
Incidence	0°	-3°	0°
Taper Ratio	1	1	0.5

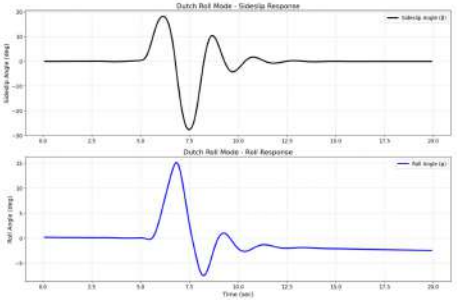
CONTROL SURFACE SIZING

Control Surfaces	% of MAC	% of Span	Deflection (deg)
Aileron	22	60	-20 to 25
Elevator	35	90	-20 to 25
Rudder	27	100	-25 to 30
Flap	30	40	0 to 40

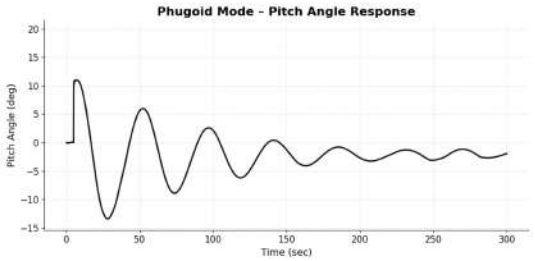
AEROMBULANCE : THE AEROEVACUATION MEDICAL AIRCRAFT



DUTCH ROLL



PHUGOID MODE



SHORT PERIOD & SPIRAL MODE

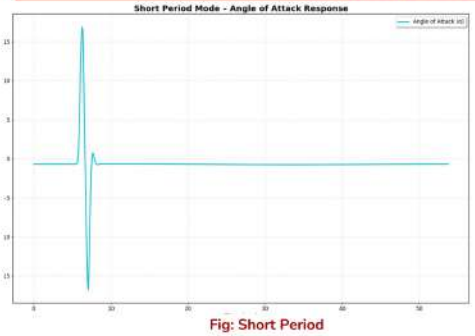


Fig: Short Period

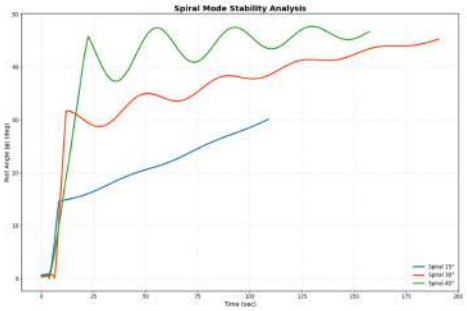
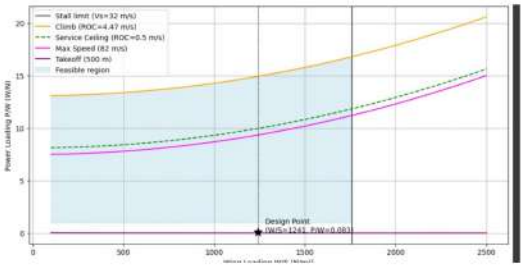
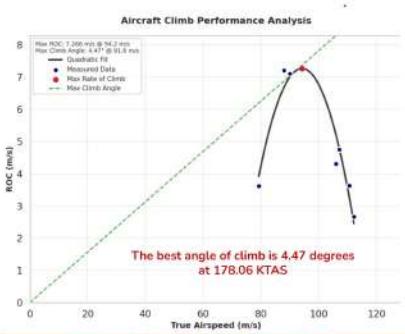


Fig: Spiral Mode

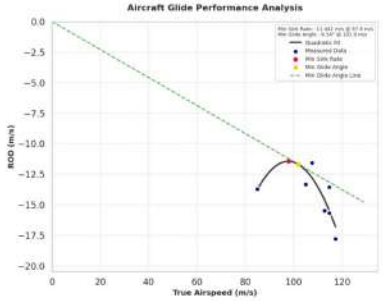
MATCHING PLOT



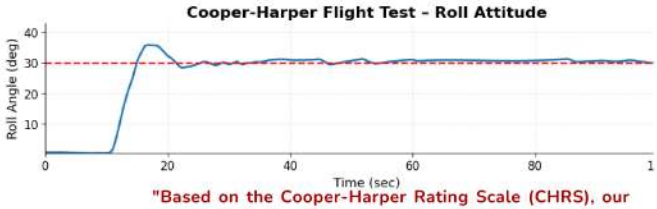
RATE OF CLIMB(ROC)



RATE OF DESCENT(ROD)

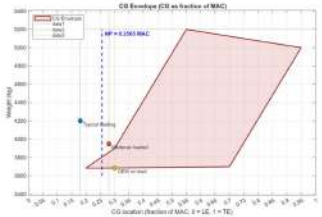


Cooper - Harper Scale

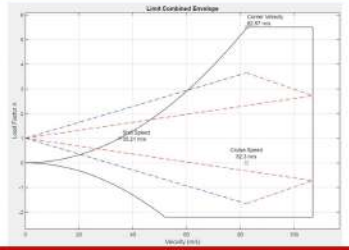


"Based on the Cooper-Harper Rating Scale (CHRS), our aeromedical aircraft demonstrates satisfactory handling characteristics and can be effectively controlled within a specified range of pitch and roll variations. The pilot is required to put in moderate effort to maintain precision. Thus, we assign a Cooper-Harper rating of 3 to our aircraft."

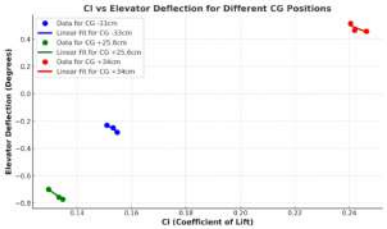
CG ENVELOPE

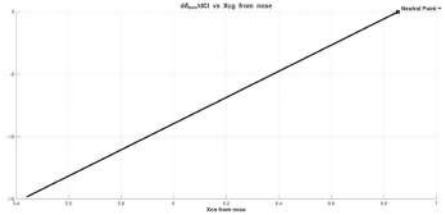


FLIGHT AND GUST ENVELOPE

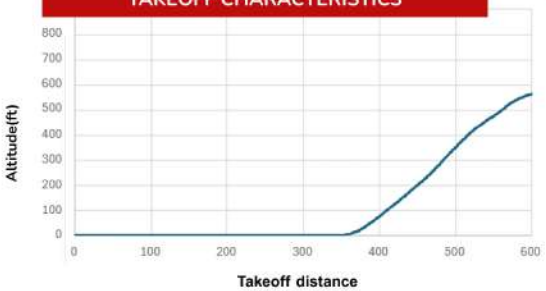


DETERMINATION OF NEUTRAL POINT

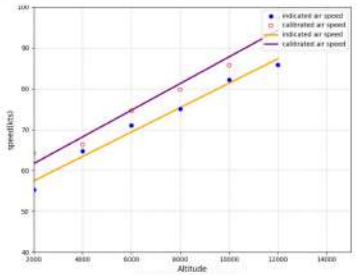




TAKEOFF CHARACTERISTICS



STALL CHARACTERCISTICS



CONCLUSION

"Preliminary flight tests of Aerombulance concluded that we achieved our design requirements of Range (1200km) and Endurance of 5.1 hrs. We were also able to perform effectively. The aircraft also exhibits good static and dynamic stability, with minor improvements required to further reduce yawing tendency and enhance handling qualities."